HELLHOUSE

Save Prapti From The Darkness

PROJECT DOCUMENTATION

A Top-Down Demon Hunter Pygame Game

Field	Details
Game Title	HELLHOUSE — Save Prapti
Genre	Top-Down Shooter / Action
Engine	Python 3.11 + Pygame 2.6
Platform	Windows / Linux / macOS
Creator	Prapti
Language	Python (single file, no OOP)
Save Files	scores.json | sessionlogs.txt

1. Project Overview

HELLHOUSE is a fast-paced top-down shooter built entirely in Python using Pygame. The player controls a character navigating a dark, multi-room house infested with demons. The mission is to fight through waves of enemies, defeat the Demon Lord and his pet Dragon who guard Prapti, and rescue her to win the game.

The entire game — map, graphics, sounds, AI, HUD, leaderboard — is written in a single **main.py** file using plain functions and dictionaries. No classes, no external assets, no image files. Everything is drawn and generated at runtime.

Metric	Value
Lines of code	~1,300
External files	0 (everything generated in code)
Save files	scores.json + sessionlogs.txt
Enemy types	3 (Grunt, Shooter, Demon Lord Boss)
Boss pets	1 (Fire-breathing Dragon)
Map tiles	7 types (floor, wall, door, window, blood, carpet, bridge)
Map size	42 x 32 tiles (2016 x 1536 px world)
Resolution	1280 x 720 px
Target FPS	60

2. Setup & Installation

2.1 Requirements

The game needs only two Python packages. Everything else is from the standard library.

Package	Version	Purpose
pygame	2.6.1	Game window, input, drawing, sound
numpy	any	Procedural sound generation (optional — game runs without it silently)

2.2 Install Dependencies

```
pip install pygame numpy
```

Or use the included requirements.txt:

```
pip install -r requirements.txt
```

2.3 Project File Structure

File	Description
main.py	Entire game — all logic, drawing, AI, HUD
requirements.txt	Two-line dependency list (pygame + numpy)
scores.json	Auto-generated on first run — persists hi-score & leaderboard
sessionlogs.txt	Auto-generated — one log line appended per game session

2.4 How to Run

```
python main.py
```

No configuration needed. The game window opens at 1280x720. The scores.json and sessionlogs.txt are created automatically in the same folder as main.py on the first run.

3. Controls

Key / Input	Action
W / Arrow Up	Move forward (up)
S / Arrow Down	Move backward (down)
A / Arrow Left	Move left
D / Arrow Right	Move right
Mouse movement	Aim weapon — character rotates to face cursor
Left mouse click	Shoot — fires bullet toward cursor position
ESC	Quit the game immediately

Note: Movement and aiming are independent. You can strafe sideways while aiming in any direction. Hold the left mouse button for continuous fire.

4. Gameplay & Features

4.1 Core Gameplay Loop

The player starts in the top-left room of the house. Waves of demons spawn throughout the map. The Demon Lord Boss and his Dragon permanently guard Prapti in the bottom section. The player must navigate through the house, killing all demons, defeat the Boss and Dragon, then walk to Prapti to rescue her.

4.2 Enemy Types

Enemy	HP	Behaviour	Score
Grunt Demon	65+	Charges player, melee attack	120+
Shooter Demon	52+	Keeps distance, fires purple projectiles	180+
Demon Lord	600	Guards Prapti, 4 shot patterns, enrages at 40% HP	3000
Fire Dragon	200	Orbits boss, breathes 3-way fire bursts	800

4.3 Surround Mechanic (Unique Feature)

When the player gets within **350 pixels of the Demon Lord**, every grunt and shooter demon on the map immediately abandons their normal AI and enters **Surround Mode**. They form a rotating formation around the player at staggered angles, completely surrounding and trapping them. Shooters also fire more frequently from their surround positions. This makes approaching the boss extremely dangerous and requires clearing demons first.

4.4 Boss Fight

The Demon Lord has four attack patterns that cycle:

- Ring burst — 8 bullets fired in all directions simultaneously
- Aimed triple — 3 fast projectiles aimed directly at player
- Spiral — 4 rotating bullets in a spiral pattern
- Cross burst — 12-way ring (unlocked when enraged at 40% HP)

The Dragon orbits the boss in a circle, breathing 3-way fire bursts every ~50 frames. Both the Boss AND the Dragon must be killed before Prapti can be rescued.

4.5 Pickups

Pickup	Effect	Drop Chance
AMMO	+20 ammo (max 60)	35% from any enemy kill
+HP	+30 health (max 120)	35% from any enemy kill
$500	+500 score	35% from any enemy kill

4.6 Win Condition

Kill the Demon Lord Boss → Kill the Fire Dragon → Walk within 80px of Prapti. The win screen appears with final score, demon kills, and the updated leaderboard.

5. Map Layout

The house map is 42 x 32 tiles (each tile is 48x48 pixels, giving a 2016x1536 world). The camera follows the player and clamps to map boundaries.

Tile Code	Type	Description
0	Floor	Walkable dark wood floor
1	Wall	Solid brick wall — blocks movement and bullets
2	Door	Decorative door — walkable passage between rooms
3	Window	Wall with window — blocks movement and bullets
4	Blood	Blood stain on floor — purely decorative
5	Carpet	Red carpet — purely decorative, walkable
6	Bridge	Wooden plank bridge — connects upper and lower sections

The map has three horizontal sections connected by two wooden bridges at columns 13 and 23. Prapti and the boss are located in the bottom section (rows 24-30).

Map Overview (schematic):

```
■■■
```

Minimap	Top-right	180x130px live map with coloured dots
Crosshair	Mouse pos	Yellow circle + cross, white centre dot

Minimap Legend:

Dot Colour	Represents
Teal / Cyan	Player position
Red	Grunt or Shooter demon
Orange	Demon Lord Boss (larger dot, pulsing)
Green	Fire Dragon
Pink	Prapti location (pulsing when close)

7. Save Files

7.1 scores.json

Created automatically in the project folder on the first completed game. Updated after every game session (death or win). Stores persistent high score, total stats, and the top-10 leaderboard sorted by score.

Example scores.json:

```
{
"hi_score": 8500,
"total_games": 12,
"total_kills": 247,
"leaderboard": [
{
"score": 8500,
"kills": 34,
"wave": 4,
"result": "WIN",
"date": "2025-03-21 14:32"
},
{
"score": 4200,
"kills": 18,
"wave": 2,
"result": "DEAD",
"date": "2025-03-21 14:10"
}
]
}
```

7.2 sessionlogs.txt

One line is appended per game session, never overwritten. Useful for tracking performance over time. Each line records timestamp, result, score, kills, wave, and session duration in seconds.

Example sessionlogs.txt:

```
[2025-03-21 14:32:05] result=WIN score=8,500 kills=34 wave=4 time=223s
[2025-03-21 14:10:12] result=DEAD score=4,200 kills=18 wave=2 time=94s
[2025-03-21 13:55:40] result=DEAD score=1,050 kills=7 wave=1 time=38s
```

Field	Type	Description
hi_score	int	All-time highest score across all sessions
total_games	int	Total number of game sessions played
total_kills	int	Total enemy kills across all sessions

leaderboard	list	Top 10 entries sorted by score descending
score	int	Score for that session
kills	int	Enemy kills in that session
wave	int	Wave reached when session ended
result	string	WIN or DEAD
date	string	Timestamp YYYY-MM-DD HH:MM

8. Code Architecture

The entire game is one file — main.py — written with plain functions and Python dictionaries. No classes are used anywhere, making the code easy to read and modify for anyone learning Python.

Section	Lines (approx)	What it does
Imports & Init	1-50	pygame, math, random, json, os setup
Colours & Fonts	51-100	All colour constants and font loading
Sound generation	101-145	numpy sine/square wave beep generator
Map definition	146-210	42x32 tile map as list of strings
Particles	211-250	Explosion/hit particle system
Screen shake	251-270	Global shake amount and offset logic
Camera	271-280	Clamped camera following player
Player	281-400	Dict, movement, shooting, drawing
Bullets	401-450	Player and enemy bullet logic
Grunt demon	451-530	Melee AI + surround mode + drawing
Shooter demon	531-620	Ranged AI + surround mode + drawing
Demon Lord (boss)	621-730	4 patterns, enrage, HP bar
Dragon	731-830	Orbit AI, fire breath, wing animation
Prapti	831-890	Rescue trigger, animated drawing
Pickups	891-940	Ammo / health / score drops
Map drawing	941-1010	Tile renderer with all 7 tile types
Minimap	1011-1060	180x130 overview map with dots
HUD	1061-1110	HP bar, ammo, score, status text
Save files	1111-1180	load/save scores.json + sessionlogs.txt
Title / Game Over / Win	1181-1280	Screen drawing with leaderboard
Main loop	1281-1400	Event loop, update all, draw all

9. Demo Notes & Screenshots Guide

The following notes describe what each screenshot of the game should show. Attach your own screenshots in place of these descriptions.

Screenshot	What to Capture	Key Elements Visible
Title Screen	Press Enter prompt with full controls li	tBlood drip animation, controls, objectives,
Gameplay 1	Player moving through rooms	Character, map, minimap, HUD bars, cros
Gameplay 2	Fighting grunt demons	Red demon enemies, bullet trails, blood pa
Gameplay 3	Surround mode active near boss	All demons circling player, flash message
Boss Fight	Engaging the Demon Lord + Dragon	Boss HP bar at top, dragon orbiting, ring
Dragon Fire	Dragon breathing fire	Green dragon, orange fire projectiles, part
Prapti Found	Reaching Prapti after boss kill	Pink glow, PRAPTI label, SAVE ME text
Win Screen	Victory screen with leaderboard	PRAPTI IS SAFE, score table, top-10 boa
Game Over	Death screen with leaderboard	YOU DIED, stats box, top-10 leaderboard
scores.json	Open file in text editor	JSON structure with leaderboard entries
sessionlogs	Open file in text editor	Log lines with timestamps and results

Tip: Run the game, play for a few minutes, die once and win once to generate real data in both scores.json and sessionlogs.txt before taking screenshots.

10. Complete Feature List

Map & World

- 42x32 tile hand-crafted house map with 7 tile types
- Three connected sections (top / middle / bottom) via wooden bridges
- Scrolling camera clamped to world boundaries
- Ambient details — blood stains, carpet, windows, doors

Player

- WASD + arrow key movement with diagonal normalisation
- Free-aim — mouse cursor controls weapon direction independently
- Ammo system — limited bullets, pickups required
- Invincibility frames after taking a hit (40 frames)
- Muzzle flash effect on every shot

Enemies

- Grunt Demon — melee charger with animated horns and claws
- Shooter Demon — ranged attacker with tentacles and three eyes
- Demon Lord Boss — 4 attack patterns, enrages at 40% HP
- Fire Dragon — orbits boss, breathes 3-way fire, charges when boss dies

- All enemies have HP bars above their heads

Surround Mechanic

- All grunts/shooters enter formation mode when player nears boss
- Rotating formation at staggered angles around player
- Increased damage and fire rate in surround mode
- Resets to normal AI when player retreats from boss area

Progression

- Wave-based enemy spawning — harder each wave
- Boss and Dragon always present — permanently guard Prapti
- New demon waves spawn when all regular enemies are cleared
- Score multiplied by wave level

Visual Effects

- Particle system — blood splats, muzzle flash, pickup sparkles
- Screen shake on hits, kills, and boss events
- Animated enemies — bobbing, spinning, tentacle movement
- Dragon wing flap animation, tail wag, fire breath glow
- Prapti rescue glow pulse effect
- Blood drip animation on title screen

HUD & UI

- HP bar (green/orange/red) with pulsing border
- Ammo display with low-ammo warning
- Pulsing score display
- Wave counter
- Boss HP bar at top of screen with enrage warning
- Live minimap (180x130) with coloured entity dots
- Custom crosshair (circle + lines + dot)
- Flash messages for events (wave start, boss kills, etc.)

Save System

- scores.json — persistent hi-score, total stats, top-10 leaderboard
- sessionlogs.txt — timestamped log of every session
- Auto-created on first run, never requires manual setup
- Leaderboard displayed on Game Over and Win screens
- Corrupt file auto-recovery — backs up and resets

Sound

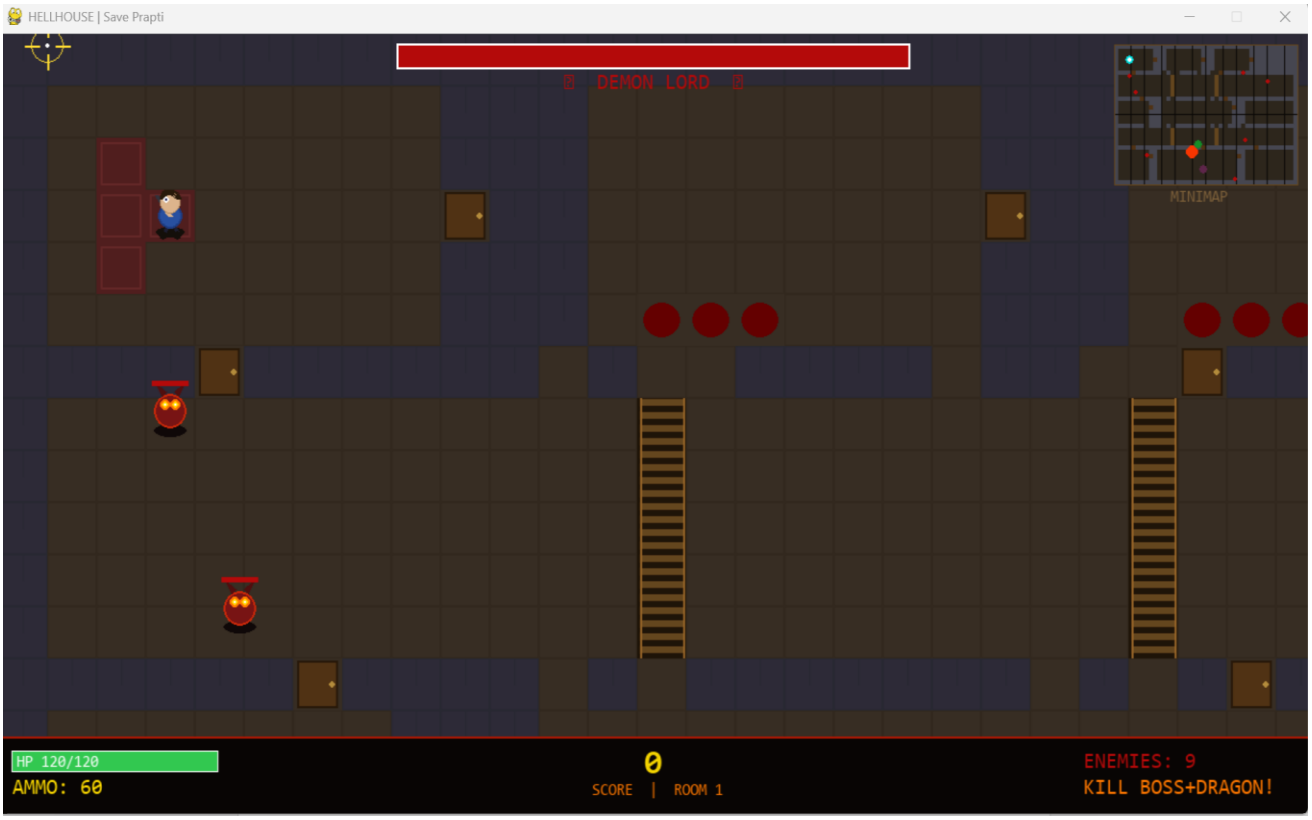
- Procedurally generated beep sounds using numpy
- Shoot, enemy fire, hit, die, pickup, rescue, roar, dragon sounds

- Graceful fallback — runs silently if numpy not installed

11. Quick Reference Card

Topic	Value / Command
Run the game	python main.py
Install dependencies	pip install pygame numpy
Player start position	Top-left room (tile 3,3)
Prapti location	Bottom section (tile 20,28)
Boss location	Bottom section (tile 20,26)
Rescue distance	80 pixels from Prapti
Surround trigger	Within 350px of boss
Player max HP	120
Player max ammo	60
Boss HP	600 (enrages at 240)
Dragon HP	200
Shoot cooldown	8 frames (~133ms at 60fps)
Bullet speed	14 px/frame (player) / 7-10 (enemies)
Hi-score file	scores.json (same folder)
Session log file	sessionlogs.txt (same folder)
Map tile size	48 x 48 pixels
World size	2016 x 1536 pixels (42x32 tiles)
Window size	1280 x 720 pixels
Target framerate	60 FPS

12 .Screen Shots



HELLHOUSE | Save Prapti
DEMON LORD
MINIMAP
PRAPTI
SAVE ME!
HP 44/120
AMMO: 47
240
SCORE | ROOM 1
ENEMIES: 7
KILL BOSS+DRAGON!

HELLHOUSE | Save Prapti
YOU DIED
Score : 240
Best : 2,140
Kills : 2
TOP 10 LEADERBOARD
Total games: 3 Total kills: 10
1. score=2,140 kills=8 wave=2 DEAD 2026-03-28 18:13
2. score=240 kills=2 wave=1 DEAD 2026-03-28 18:16
3. score=0 kills=0 wave=1 DEAD 2026-03-27 22:50
ENTER – retry ESC – quit